11. The End of Infinity

When we take quantum gravity into account, the infinite compression of the universe into a single, infinitely small point predicted by general relativity at the Big Bang disappears. Quantum gravity is the discovery that no infinitely small point exists. There is a lower limit to the divisibility of space. The universe cannot be smaller than the Planck scale, because nothing exists which is smaller than the Planck scale.

If we ignore quantum mechanics, we ignore the existence of this lower limit. The pathological situations predicted by general relativity, where the theory gives infinite quantities, are called singularities. Quantum gravity places a limit to infinity, and 'cures' the pathological singularities of general relativity.

The same happens at the centre of black holes: the singularity that classic general relativity anticipated disappears as soon as we take quantum gravity into account.

There is another case, of a different kind, in which quantum gravity places a limit to the infinite, and it regards forces such as electromagnetism. Quantum field theory, started by Dirac and completed in the 1950s by Feynman and his colleagues, describes these forces well but is full of mathematical absurdities. When we use it to compute physical processes, we often obtain results which are infinite, and mean nothing. They are called divergences. The divergences are then eliminated with calculations, using a baroque technical procedure which leads to finite final results. In practice, it works, and the numbers, in the end, come out right; they reproduce the experimental measurements. But why must the theory go via the infinite to arrive at reasonable numbers?

In the last years of his life, Dirac was very unhappy with the infinities in his theory and felt that, all things considered, his objective of truly understanding how things worked was not achieved. Dirac loved conceptual clarity, even if what was clarity to him was not always clarity to others. But infinities do not make for clarity.

But the infinities of quantum field theory follow from an assumption at the basis of the theory: the infinite divisibility of space. For example, to calculate the probabilities of a process, we sum up – as Feynman has taught us – all of the ways in which the process could unfold, and these are infinite, because